

Yujie Zhao

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EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China	Sep 2024 - Present
M.Eng. in Electronic and Information Engineering	
Advisor: Prof. Jiabei Zeng	
GPA: 3.97/4.0	
University of Chinese Academy of Sciences (UCAS), Beijing, China	Sep 2020 - Jun 2024
B.E. in Computer Science and Technology	
GPA: 3.85/4.0, Rank: 18/126	
University of California, Berkeley, CA, USA	Aug 2023 - Dec 2023
Visiting Student in Electrical Engineering and Computer Sciences	
GPA: 3.9/4.0	

PUBLICATION

- [1] **LangSplatV2: High-dimensional 3D Language Gaussian Splatting with 450+ FPS**
Wanhua Li*, Yujie Zhao*, Minghan Qin, Yang Liu, Yuanhao Cai, Chuang Gan, Hanspeter Pfister (*first author as a student*)
In submission NeurIPS 2025
- [2] **TwinAligner: Visual and Physical Real2Sim2Real All-in-one for Robotic Manipulation**
Hongwei Fan*, Hang Dai*, Jiayao Zhang*, Jinzhou Li, Qiyang Yan, Yujie Zhao, Yuxuan Lai, Hao Tang, Hao Dong
In submission CoRL 2025
- [3] **Pose-Robust Calibration Strategy for Point-of-Gaze Estimation on Mobile Phones**
Yujie Zhao, Jiabei Zeng, Shiguang Shan
British Machine Vision Conference (BMVC) 2025
- [4] **Multi-view 3D Gaze Target Detection**
Yunjia Sun*, Yujie Zhao*, Jiabei Zeng, Shiguang Shan (*first author as a student*)
In submission Pattern Recognition (PR)

RESEARCH EXPERIENCES

3D Reconstruction and Robot Manipulation PKU-AgiBot Lab, Peking University	April 2025 – Present
Advisor: Hao Dong, Professor	
➤ Utilized advanced 3D reconstruction techniques such as PGSR and NeuS to achieve high-fidelity reconstruction of real-world objects and scenes. The reconstructed assets were rendered in a simulator, significantly reducing the sim-to-real visual gap. ➤ Reproduced state-of-the-art 3D models including MAST3R and VGGT, and fine-tuned them using multi-view manipulation data collected from humanoid robots. This enabled real-time 3D scene reconstruction and camera pose estimation in robot manipulation environments.	
Real-Time 3D Language Gaussian Splatting Visual Computing Group, Harvard University	July 2024 – Mar 2025
Advisor: Hanspeter Pfister, Professor and Wanhua Li, Postdoc	
➤ Made analysis on each step of the query process of LangSplat, which is an advanced framework for open-vocabulary querying withing 3D spaces and found the primary bottleneck of LangSplat's inference speed is the heavy weight decoder. ➤ Proposed to model a 3D sparse field instead of assigning each Gaussian with a language feature, eliminating the decoder bottleneck of inference speed, and better directly modeling high-dimensional language feature field. ➤ Proposed efficient sparse coefficient splatting with CUDA optimization, decoupling the render speed from the render dimension, and achieving high-dimensional feature splatting at $42 \times$ speed and open-vocabulary querying at $30 \times$ speed compared to LangSplat respectively. ➤ Conducted experiments and demonstrated our method not only achieves the SOTA performance but is also significant faster.	
Multi-view 3D Gaze Target Detection Institute of Computing Technology, CAS	Dec 2024 – Mar 2025

Advisor: Jiabei Zeng, Associate Professor

- Designed the collection process of the multi-view gaze target detection dataset and built the scene for dataset collection.
- Collected data and constructed the first multi-view 3D gaze target detection benchmark.
- Proposed an attention-based model for extracting and fusing multi-view features, outperforming traditional 2D gaze target detection methods on various evaluation metrics of the benchmark.

Self-supervised Gaze Point Estimation | Institute of Computing Technology, CAS

Dec 2023 – May 2024

Advisor: Prof. Jiabei Zeng, Associate Professor

- Collected and organized a large number of gaze-related datasets and built a unified data preprocessing pipeline.
- Designed local and global masking strategies, enabling the MAE model to learn image features of different granularities that are useful for gaze point estimation from unlabeled data.
- Performed supervised fine-tuning on the pre-trained model, achieving excellent performance on the test set.

Gaze Point Estimation Dataset Collection | Institute of Computing Technology, CAS

Jan 2023 - Jun 2023

Advisor: Prof. Jiabei Zeng, Associate Professor

- Designed a data collection process that can collect gaze data from images with diverse poses, resolving a lack of public gaze point estimation datasets and a lack of diversity in subjects' postures.
- Added voice prompts and psychological techniques into the collection process to ensure the accuracy of labels.
- Implemented the designed data collection process into Android applications using Android Studio, so that the data collection tools can be easily deployed on different mobile devices.
- Recruited volunteers and guided them in conducting data collection.
- Preprocessed the collected data and formed a dataset, which, to the best of our knowledge, is the first gaze dataset that consists continuously changing postures.

HONORS

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| ➤ University Academic Scholarship at University of Chinese Academy of Sciences (UCAS) (20%) | Nov 2021 |
| ➤ University Academic Scholarship at University of Chinese Academy of Sciences (UCAS) (20%) | Nov 2022 |
| ➤ Merit Student at University of Chinese Academy of Sciences (UCAS) (25%) | Jun 2023 |
| ➤ University Academic Scholarship at University of Chinese Academy of Sciences (UCAS) (20%) | Nov 2023 |
| ➤ Merit Student at University of Chinese Academy of Sciences (UCAS) (25%) | Jun 2024 |

SKILLS

- **Programming Language:** C, C++, Python, Java, Verilog
- **Operating System:** Linux, Windows, MacOS
- **Tools:** SSH and other tools to efficiently operate and manage servers for computational tasks.
- **Standard English Tests:** IELTS 7